

**SIMATS ENGINEERING
ASSESSMENT TOOL 2**

COURSE CODE & NAME: CSA16 / Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.

Assessment Tool Weightage: 40%

QUESTIONS: 05

Total Marks:50

Annexure A

COMPARITIVE ANALYSIS- Industry Problem Solving Task

Criteria	Understanding of Industry Problem (2)	Application of Engineering Concepts (2.5)	Solution Design (2.5)	Use of Tools (2)	Analysis (1)	Total (10)
<i>Weightage</i>	20%	25%	25%	20%	10%	100%
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of Conduct:

I Ajay I (192421147) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *Ajay J*

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Annexure A

Q1. Dataset: Supermarket Purchases

Customer	Visits/Month	Avg Spend (₹)	Items Bought
C1	10	2000	15
C2	4	5000	8
C3	12	1800	18
C4	3	6000	6
C5	11	2100	16

Question:

Using the above dataset, propose a clustering approach to segment customers and explain how it supports targeted marketing. *(BL3 – 7 Marks)*

Q2. Dataset: Credit Card Transactions

Transaction	Amount (₹)	Location	Time (hrs)
T1	200	Chennai	10
T2	5000	Delhi	2
T3	150	Chennai	11
T4	7000	Mumbai	1
T5	180	Chennai	12

Question:

Design a clustering-based solution using this dataset to detect unusual or fraudulent transactions. *(BL4 – 7 Marks)*

Q3. Dataset: Telecom Usage

User	Call Duration (min)	Data Usage (GB)	Recharge (₹)
U1	300	5	200
U2	100	2	100
U3	350	6	220
U4	80	1	90
U5	320	5.5	210

Question:

Explain how clustering can be applied to identify churn-prone users and improve customer retention strategies. *(BL4 – 7 Marks)*

Q4. Dataset: E-commerce Behaviour

User	Product Views	Clicks	Purchases
U1	50	20	5
U2	30	10	2
U3	60	25	6
U4	20	8	1
U5	55	22	5

Question:

Suggest a clustering technique to group users and explain how it improves recommendation systems. *(BL3 – 7 Marks)*

Q5. Dataset: Traffic Data

Location	Vehicle Count	Avg Speed (km/h)	Delay (min)
L1	200	30	10
L2	500	15	25
L3	220	28	12
L4	600	10	30
L5	210	32	9

Question:

Design a clustering-based solution to identify traffic congestion zones and explain how it helps in traffic management. (BL4 – 7 Marks)

ANSWERS:

Q1. Supermarket Purchases – Customer Segmentation

Clustering approach: Use **K-Means clustering**.

The attributes are:

- Customer visits/month
- Average spending
- Items bought

First, normalize the numerical values because their scales are different. Then apply K-Means with, for example, **K = 2 or 3** clusters.

For this dataset:

- **High-value frequent customers:** C1, C3, C5
- **Low-frequency high-spending customers:** C2, C4

The clusters can help the supermarket understand different customer behaviours.

Targeted marketing:

- Frequent customers → loyalty rewards and personalized offers.
- High-spending customers → premium products and special discounts.
- Low-frequency customers → coupons and promotional campaigns to encourage visits.

Thus, clustering helps the supermarket provide **different marketing strategies for different customer segments**.

Q2. Credit Card Transactions – Fraud Detection

Clustering approach: Use **K-Means clustering** or **DBSCAN**.

The features are:

- Transaction amount
- Location
- Transaction time

Numerical features should first be normalized. Location can be converted into suitable numerical/encoded values.

Normal transactions are expected to form groups based on their amount, location and time. Transactions that are **far away from normal clusters** can be considered suspicious.

For example:

- T1, T3 and T5 have small transaction amounts and similar locations/times.
- T2 and T4 have very large amounts and different locations/times.

Therefore, T2 and T4 can be investigated as potentially unusual transactions.

Fraud detection process:

Transaction Data → Preprocessing → Clustering → Identify Outliers → Fraud Investigation

This approach helps banks automatically identify unusual transaction patterns and reduce fraudulent activities.

Q3. Telecom Usage – Churn Prediction

Clustering approach: Apply **K-Means clustering** to group users according to their usage behaviour.

Features:

- Call duration
- Data usage
- Recharge amount

Possible groups are:

- **High-usage users:** U1, U3, U5
- **Low-usage users:** U2, U4

Users with **low call usage, low data usage and low recharge activity** can be considered more likely to reduce or stop using the service.

Retention strategies:

- Give low-usage customers special recharge offers.
- Provide additional data benefits.
- Send personalized plans.
- Offer discounts to encourage continued usage.
- Provide loyalty rewards to active customers.

Therefore, clustering helps telecom companies identify customer groups and develop **targeted retention strategies**.

Q4. E-Commerce Behaviour – Recommendation System

Clustering technique: Use **K-Means clustering**.

Features:

- Product views
- Clicks
- Purchases

Users with similar behaviour are grouped together.

For example:

- **Highly active buyers:** U1, U3, U5
- **Moderately active users:** U2
- **Low-activity users:** U4

The recommendation system can then recommend products based on the behaviour of users in the same cluster.

Example:

If U1, U3 and U5 frequently view, click and purchase similar products, products purchased by one user can be recommended to the others.

Process:

User Behaviour → Clustering → User Segments → Identify Preferences → Personalized Recommendations

Thus, clustering improves recommendation systems by providing **personalized product suggestions**, increasing customer engagement and potential sales.

Q5. Traffic Data – Congestion Zones

Clustering approach: Use **K-Means clustering** or **DBSCAN**.

Features:

- Vehicle count
- Average speed
- Delay

The traffic locations can be grouped according to their congestion characteristics.

From the dataset:

- **High congestion:** L2 and L4
- **Moderate congestion:** L1 and L3
- **Low congestion:** L5

L2 and L4 have **high vehicle counts, low average speeds and high delays**, indicating severe congestion.

Traffic management actions:

- Adjust traffic signal timings.
- Deploy traffic police to congested locations.
- Provide alternative routes.
- Implement intelligent traffic management systems.
- Monitor congestion in real time.

Process:

Traffic Data → Preprocessing → Clustering → Congestion Zones → Traffic Management

Thus, clustering helps authorities identify congestion-prone areas and take appropriate actions to **improve traffic flow and reduce delays**.